Cargo las librerias

In [1]:
```
import pandas as pd
import numpy as np

pd.set_option('display.max_columns', None)
```

Cargo los set de datos

In [8]:
```
df_personas = pd.read_csv(
    "H224.csv",
    sep=";",
    low_memory=False
)

df_condiciones = pd.read_csv(
    "h222.csv",
    sep=";",
    low_memory=False
)

df_medicamentos = pd.read_csv(
    "h229a.csv",
    sep=";",
    low_memory=False
)
```

verifico

In [4]:
```
df_personas.shape
```

Out[4]:
```
(27805, 1451)
```

In [5]:
```
df_condiciones.shape
```

Out[5]:
```
(80802, 30)
```

In [6]:
```
df_medicamentos.shape
```

Out[6]:
```
(284375, 66)
```

Selecciono Variables utiles

Estas variables existen en H224 y son relevantes para el proyecto

In [9]:
```
variables_personas = [
    "DUPERSID",
    "AGE20X",
    "SEX",
    "MARRY20X",
    "EDUCYR",
```

```
    "POVCAT20",
    "EMPST53",
    "INSCOV20",
    "MNHLTH53",
    "RTHLTH53",
    "RACETHX"
]

df_personas = df_personas[variables_personas]
```

Exploracion

In [8]:
```
df_personas.head()
```

Out[8]:

	DUPERSID	AGE20X	SEX	MARRY20X	EDUCYR	POVCAT20	EMPST53	INSCOV20
0	2320005101	73	2	1	14	2.0	4.0	2.0
1	2320005102	84	1	1	12	2.0	4.0	2.0
2	2320006101	47	2	3	12	3.0	4.0	3.0
3	2320006102	22	1	5	11	1.0	1.0	3.0
4	2320006103	21	1	5	11	3.0	4.0	2.0

In [9]:
```
df_personas.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6642 entries, 0 to 6641
Data columns (total 11 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   DUPERSID  6642 non-null   int64
 1   AGE20X    6642 non-null   int64
 2   SEX       6642 non-null   int64
 3   MARRY20X  6642 non-null   int64
 4   EDUCYR    6642 non-null   int64
 5   POVCAT20  6641 non-null   float64
 6   EMPST53   6641 non-null   float64
 7   INSCOV20  6641 non-null   float64
 8   MNHLTH53  6641 non-null   float64
 9   RTHLTH53  6641 non-null   float64
 10  RACETHX   6642 non-null   int64
dtypes: float64(5), int64(6)
memory usage: 570.9 KB
```

condiciones medicas

Cantidad de condiciones por persona.

In [10]:
```
df_condiciones_persona = (
    df_condiciones
    .groupby("DUPERSID")
    .agg(
        cantidad_condiciones=("DUPERSID","count")
    )
```

```
    .reset_index()
)

df_condiciones_persona.head()
```

Out[10]:

	DUPERSID	cantidad_condiciones
0	2320005101	1
1	2320005102	2
2	2320006102	1
3	2320006103	1
4	2320012102	5

LIMPIEZA DE MEDICAMENTOS

In [13]:

```
df_medicamentos["RXDRGNAM"] = (
    df_medicamentos["RXDRGNAM"]
    .astype(str)
    .str.upper()
    .str.strip()
)
```

CLASIFICACIÓN DE PSICOFÁRMACOS

In [14]:

```
benzodiacepinas = [
    "ALPRAZOLAM",
    "CLONAZEPAM",
    "DIAZEPAM",
    "LORAZEPAM",
    "TEMAZEPAM",
    "MIDAZOLAM"
]

antidepresivos = [
    "FLUOXETINE",
    "SERTRALINE",
    "PAROXETINE",
    "ESCITALOPRAM",
    "CITALOPRAM",
    "VENLAFAXINE",
    "DULOXETINE",
    "BUPROPION",
    "AMITRIPTYLINE"
]

hipnoticos = [
    "ZOLPIDEM",
    "ZALEPLON",
    "ESZOPICLONE"
]

ansioliticos = [
    "BUSPIRONE",
```

```
    "HYDROXYZINE"
]
```

V. Derivadas

In [15]:
```
df_medicamentos["benzo"] = (
    df_medicamentos["RXDRGNAM"]
    .isin(benzodiacepinas)
    .astype(int)
)

df_medicamentos["antidepresivo"] = (
    df_medicamentos["RXDRGNAM"]
    .isin(antidepresivos)
    .astype(int)
)

df_medicamentos["hipnotico"] = (
    df_medicamentos["RXDRGNAM"]
    .isin(hipnoticos)
    .astype(int)
)

df_medicamentos["ansiolitico"] = (
    df_medicamentos["RXDRGNAM"]
    .isin(ansioliticos)
    .astype(int)
)
```

AGR. DE PSICOFÁRMACOS

In [16]:
```
df_psicofarmacos = (
    df_medicamentos
    .groupby("DUPERSID")
    .agg(
        recetas_totales=("RXDRGNAM","count"),
        benzodiacepinas=("benzo","sum"),
        antidepresivos=("antidepresivo","sum"),
        hipnoticos=("hipnotico","sum"),
        ansioliticos=("ansiolitico","sum")
    )
    .reset_index()
)

df_psicofarmacos["total_psicofarmacos"] = (
    df_psicofarmacos["benzodiacepinas"]
    + df_psicofarmacos["antidepresivos"]
    + df_psicofarmacos["hipnoticos"]
    + df_psicofarmacos["ansioliticos"]
)
```

union

In [17]:
```
df_final = (
    df_personas
```

```
    .merge(
        df_condiciones_persona,
        on="DUPERSID",
        how="left"
    )
    .merge(
        df_psicofarmacos,
        on="DUPERSID",
        how="left"
    )
)
```

limpio nulos

In [18]:
```
df_final.fillna(0, inplace=True)
```

verifico

In [19]:
```
df_final.shape
```

Out[19]: (27805, 18)

In [20]:
```
df_final.head()
```

Out[20]:

	DUPERSID	AGE20X	SEX	MARRY20X	EDUCYR	POVCAT20	EMPST53	INSCOV20
0	2320005101	73	2	1	14	2	4	2
1	2320005102	84	1	1	12	2	4	2
2	2320006101	47	2	3	12	3	4	3
3	2320006102	22	1	5	11	1	1	3
4	2320006103	21	1	5	11	3	4	2

In [21]:
```
df_final.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27805 entries, 0 to 27804
Data columns (total 18 columns):
 #   Column                 Non-Null Count  Dtype  
---  ------                 --------------  -----  
 0   DUPERSID               27805 non-null  int64  
 1   AGE20X                 27805 non-null  int64  
 2   SEX                    27805 non-null  int64  
 3   MARRY20X               27805 non-null  int64  
 4   EDUCYR                 27805 non-null  int64  
 5   POVCAT20               27805 non-null  int64  
 6   EMPST53                27805 non-null  int64  
 7   INSCOV20               27805 non-null  int64  
 8   MNHLTH53               27805 non-null  int64  
 9   RTHLTH53               27805 non-null  int64  
 10  RACETHX                27805 non-null  int64  
 11  cantidad_condiciones   27805 non-null  float64
 12  recetas_totales        27805 non-null  float64
 13  benzodiacepinas        27805 non-null  float64
 14  antidepresivos         27805 non-null  float64
 15  hipnoticos             27805 non-null  float64
 16  ansioliticos           27805 non-null  float64
 17  total_psicofarmacos    27805 non-null  float64
dtypes: float64(7), int64(11)
memory usage: 3.8 MB
```

LIsto , exporto

In [22]:
```
df_final.to_csv(
    "dataset_psicofarmacos_final.csv",
    index=False
)
```